ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** P and R are magnets, Q is magnetic material	Repulsion is seen only between magnets.
2	**(a)** iron is magnetic and copper is not	Only iron is attracted.
3	**(d)** always in the same direction with the same pole of a magnet	Opposite pole forms at the lifting end.
4	**(b)** south-seeking pole	The end that points south is the south-seeking pole.
5	**(a)** iron only	Only iron is magnetic.
6	**(d)** B's north repels A's north	Like poles repel.
7	**(c)** repel	N facing N.
8	**(c)** Earth's magnetism slowly magnetises it	Induction by Earth's field.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The Earth's magnetic field acts on the suspended magnet and aligns it, so R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but R does not explain why iron is attracted.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The steel table disturbs the needle (1); Move it to a wooden or non-magnetic surface (1).	1+1
12	No (1); Heat weakens the magnet (1).	1+1
13	A pin can be moved through paper by a magnet (1); Non-magnetic sheets do not block force (1).	1+1
14	Nails lose temporary magnetism (1); Soft iron is not a permanent magnet (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Roll bar in filings (1); Maximum cling near the ends (1); Little in the middle (1).	1+1+1
16	Permanent magnet is present (1); A coil moves in its field (1); Produces sound (1).	1+1+1
17	Iron nails are magnetic (1); Brass screws are not magnetic (1); The magnet picks up only the nails (1).	1+1+1
	OR: Gave direction at sea (1); Not dependent on stars (1); Safer voyages (1).	1+1+1
18	Hang a bar magnet freely (1); It settles N-S (1); Repeated turns give same result (1).	1+1+1
	OR: Steel can carry magnetism (1); Nail may be pulled less (1); Magnetic materials change the field (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) North (and south) direction (1). (ii) North pole and south pole (1). (iii) The magnet's poles attracted or repelled the needle, which is itself a small magnet (1). (iv) Its north-pointing needle shows north; then south, east and west can be found (1), so travellers can find the way even where there are no landmarks (1).	1+1+1+2
20	Test each pair for attraction (1); Copper is attracted by none (1); The magnet may repel another bar's end (1); Iron is attracted by the magnet but not repelled (1); All three identified (1).	1+1+1+1+1
	OR: Heat, impact and time (1); Poor storage (1); Keepers at the poles (1); Pair with unlike poles side by side (1); Cool dry box (1).	1+1+1+1+1

Mistakes to avoid:

- Thinking everything metallic is magnetic; copper and aluminium are metals but **not** magnetic.
- Using attraction alone as the test for a magnet (iron is attracted too); the sure test is **repulsion**.
- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the pin acts as a temporary magnet	Induced magnetism.
2	**(c)** flat on the palm, away from iron objects	Keep it level and away from magnets.
3	**(c)** the first nail became temporarily magnetised	Induced magnetism.
4	**(a)** stroking in one direction repeatedly	Consistent direction aligns.
5	**(b)** the pin moves toward a magnet held at a distance	It acts across a gap.
6	**(d)** were temporary magnets	Iron loses magnetism easily.
7	**(d)** pull decreases rapidly	Pull weakens with distance.
8	**(c)** repulsion between like poles	Like poles repel.
9	**(d)** A is false but R is true.	A is false: like poles repel. R is true.
10	**(d)** A is false but R is true.	A is false, because wood is non-magnetic; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To check each other's readings (1); Iron in the ship may disturb one (1).	1+1
12	Nails are made of iron, which is magnetic (1); Wood is not a magnetic material (1).	1+1
13	See if it keeps attracting pins after removing the magnet (1); A permanent magnet will (1).	1+1
14	Doctors use a magnet (1); It pulls out iron fragments (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Needle is deflected (1); Aligns along magnet's field (1); Returns when magnet is removed (1).	1+1+1
16	Bring a magnet (1); Only iron sticks (1); Other two do not (1).	1+1+1
17	They may swallow small magnets (1); Magnets can pinch skin (1); They harm gadgets (1).	1+1+1
	OR: Magnets attract steel (1); Hold papers by force (1); No glue needed (1).	1+1+1
18	To prevent rust (1); To protect from damage (1); To give better look (1).	1+1+1
	OR: Magnet pulls pin up (1); Gravity pulls down (1); Pin rises if magnetic force is larger (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Iron key and steel pin (1). (ii) Any two of copper coin, plastic scale, glass marble (1). (iii) Magnetic materials (1). (iv) Move a magnet over the mixture (1); iron nails stick to it and the non-magnetic sand stays behind (1).	1+1+1+2
20	Home: fridge door (1); Home: speakers (1); Hospital: scanning devices (1); Industry: separating scrap (1); Transport: motors (1).	1+1+1+1+1
	OR: Compass points north (1); Freely hung magnet aligns (1); Iron objects acquire magnetism (1); Animals navigate by it (1); Evidence is consistent (1).	1+1+1+1+1

Mistakes to avoid:

- Thinking everything metallic is magnetic; copper and aluminium are metals but **not** magnetic.
- Using attraction alone as the test for a magnet (iron is attracted too); the sure test is **repulsion**.
- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** turn away	Like poles repel.
2	**(c)** keeping it with a keeper	Keepers protect.
3	**(c)** away from iron and magnets	Avoid disturbance.
4	**(d)** steel clip, iron nail	Only magnetic items are held.
5	**(a)** suspend it and see which end points north	Free suspension.
6	**(b)** a loudspeaker	It contains a magnet.
7	**(c)** 3	Each piece is a magnet with two poles.
8	**(c)** a magnet	Only a magnet has both poles.
9	**(a)** Both A and R are true and R is the correct explanation of A.	More filings stick at the poles because the attraction is strongest there.
10	**(c)** A is true but R is false.	A is true: each piece gets both poles. R is false, poles cannot be isolated.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Prevents unwanted sticking (1); Prevents loss of strength (1).	1+1
12	Cutting a magnet gives two magnets (1); Each has N and S poles (1).	1+1
13	They are non-magnetic (1); They do not disturb the magnets (1).	1+1
14	It attracts all pins at once (1); It saves time (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It weakens (1); Alignment is disturbed (1); Handle gently (1).	1+1+1
16	Knife holders (1); Spice tins (1); Fridge doors (1).	1+1+1
17	Heat a weak magnet gently (1); Test with pins before and after (1); It picks fewer pins (1).	1+1+1
	OR: Bring near pins (1); Pins stick if magnetised (1); Check poles with compass (1).	1+1+1
18	Needle deflects strongly (1); May reverse (1); Needle may lose accuracy (1).	1+1+1
	OR: Steel is attracted by the magnet (1); The magnet turns towards the table (1); It gives a wrong direction (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) They repel (1). (ii) They attract (1). (iii) The poles (1). (iv) They will repel each other and the suspended magnet turns away (1) because like poles repel (1).	1+1+1+2
20	Needle is a tiny magnet (1); Earth is like a giant magnet (1); Needle aligns with Earth's field (1); Gives direction (1); Used in ships and planes (1).	1+1+1+1+1
	OR: Temporary: soft iron near magnet (1); Permanent: steel bar magnet (1); Natural: lodestone (1); Differences in retention (1); Uses differ (1).	1+1+1+1+1

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